

Overview

Patient Retention Rate 60.0%	Total Patients 5	Retained Patients 3
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Executive Summary
This dashboard provides insights into **patient retention** and **marketing effectiveness** for prescription renewals.

- Key Business Questions:**
- Patient Retention:** What percentage of patients renew their prescriptions, and how does this trend over time?
 - Marketing Attribution:** Which marketing touchpoints drive prescription renewals?

- Navigation:**
- Patient Retention Analysis** – Detailed monthly retention trends
 - Marketing Attribution** – Campaign and channel performance analysis

Patient Retention Analysis

Understanding Patient Retention

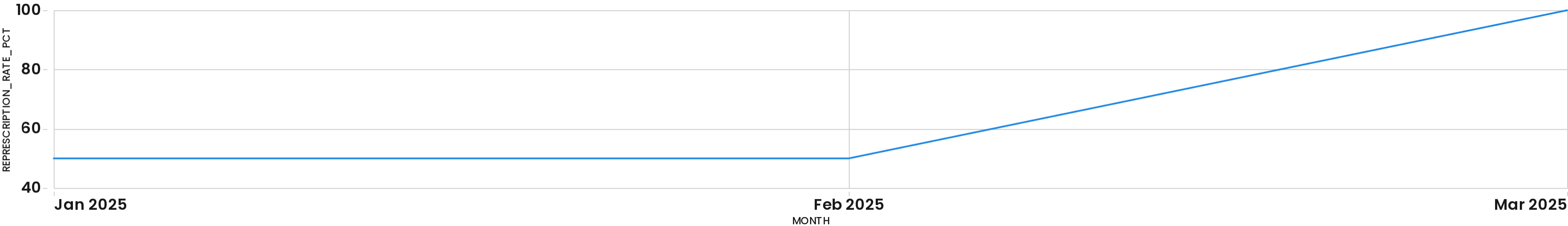
Business Context: Patient retention measures how many patients continue their prescribed treatment by getting re-prescriptions. High retention indicates effective treatment outcomes and patient satisfaction.

Metrics:

- **Retention Rate** = (Patients who renewed ÷ Total patients) × 100%
- Calculated at the **patient level** (not prescription level) to avoid double-counting
- Tracked monthly to identify seasonal trends and program effectiveness

Monthly Patient Retention Rate

Percentage of patients who renewed their prescription each month



Retention Performance

TOTAL PATIENTS	RETAINED PATIENTS	RETENTION RATE %
5	3	60.0%

Marketing Attribution

Business Question

"Which touchpoint contributes the most to re-prescription?"

Analytical Approach: Last-Touch Attribution

Methodology

We use a **last-touch attribution model** that credits the final marketing interaction before a patient renews their prescription:

- 1. **Identify renewal events:** Select all prescriptions where `represcription = TRUE`
- 2. **Find eligible touchpoints:** For each renewal, identify all touchpoints between `prescription_start` and `represcription_date`
- 3. **Attribute to last touchpoint:** Use window function to rank touchpoints by date (descending) and credit the most recent one
- 4. **Aggregate results:** Count distinct patients attributed to each touchpoint type and calculate percentage

SQL Logic

```
ROW_NUMBER() OVER (
  PARTITION BY patient_id, prescription_id
  ORDER BY touchpoint_date DESC -- Last touch!
) = 1
```

Why Last-Touch?

- Identifies the **immediate conversion driver** before renewal
- Answers: "What was the final nudge that drove the patient to renew?"
- Common in performance marketing to optimize conversion tactics

Key Findings

Based on current data:

- **CHECK_IN_CALL** (66.7%) – Phone calls are the dominant last touchpoint
- **EMAIL_NUDGE** (33.3%) – Email reminders also drive conversions
- Average timing: 65-75 days after initial prescription

Limitations & Caveats

1. Correlation ≠ Causation

Issue: The last touchpoint may have occurred *after* the patient decided to renew.

- A patient might schedule a renewal, then receive a routine check-in call
- The call didn't cause the renewal—it just happened to be last

2. Recency Bias

Issue: Over-credits the final interaction, ignores earlier influential touchpoints.

- Example: An educational email campaign (first-touch) might have been more influential than the final reminder call, but gets zero credit

3. Selection Bias

Issue: Only analyzes patients who renewed—no comparison group.

- We don't know: Did non-renewed patients receive different touchpoints?
- Without this, we can't say touchpoints *caused* renewals

4. Missing Confounders

Issue: Doesn't control for patient characteristics that drive renewal.

- Factors not measured: disease severity, medication effectiveness, doctor relationship, insurance coverage, socioeconomic status
- High-risk patients may naturally renew regardless of touchpoints

5. Multi-Touch Journey Ignored

Issue: Patients typically interact multiple times across channels.

- Attributing 100% credit to one touchpoint oversimplifies the customer journey
- Alternative models (linear, time-decay, position-based) distribute credit more fairly

6. No Counterfactual

Issue: We don't know what would have happened without touchpoints.

- Would patients have renewed anyway?
- Requires A/B testing: randomly assign patients to "touchpoint" vs. "no touchpoint" groups

Recommendations for Improvement

Immediate (use existing data)

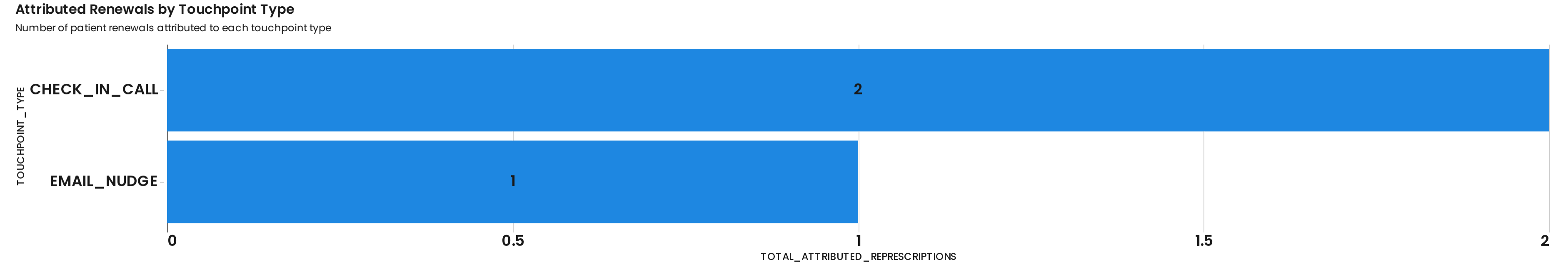
- 1. **Compare with first-touch attribution** (already in `02_gold.touchpoint_attribution_first_touch`)
- 2. **Analyze non-renewals:** Which touchpoints did patients who *didn't* renew receive?
- 3. **Segment by patient characteristics:** Does attribution differ by age, insurance type, or doctor?

Medium-term (additional analysis)

- 1. **Multi-touch attribution:** Test time-decay or position-based models
- 2. **Propensity score matching:** Match similar patients who renewed vs. didn't renew, control for confounders
- 3. **Survival analysis:** Model time-to-renewal with touchpoints as time-varying covariates

Long-term (experimentation)

- 1. **Randomized controlled trial:** Randomly assign patients to receive/not receive specific touchpoints
- 2. **Holdout testing:** Create control groups that receive no marketing touchpoints
- 3. **Incremental lift testing:** Measure the causal lift of each touchpoint type



Detailed Attribution Breakdown					
Touchpoint	Channel	Outcome	Renewals	Attribution %	Avg Days to Renewal
CHECK_IN_CALL	PHONE	completed	2	66.7%	65
EMAIL_NUDGE	EMAIL	clicked	1	33.3%	75